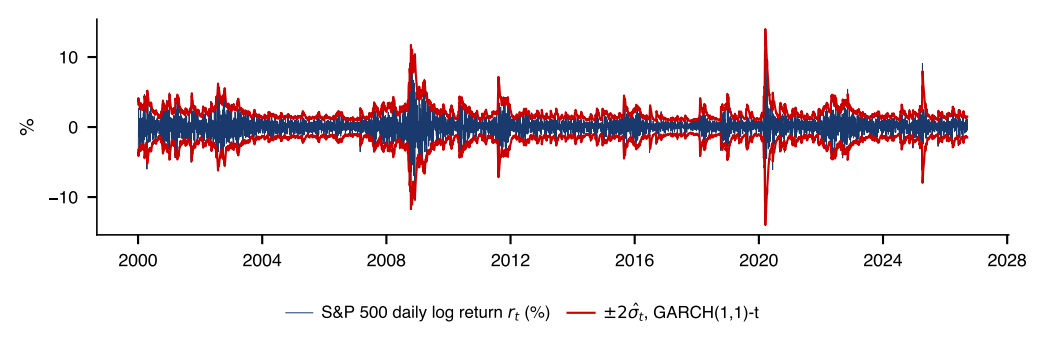

%
10
0
−10
2000
2004
2008
2012
2016
2020
2024
2028
S&P 500 daily log return r_t (%)
$\pm 2\hat{\sigma}_t$, GARCH(1,1)-t